



# ENVIRONMENTAL MONITORING REPORT

Report Type: DAILY · Generated: 4/6/2026, 2:26:26 PM

## 1. Executive Summary

This report presents environmental monitoring data collected by the EBA System over the period 3/28/2026 – 4/6/2026. A total of 108 records were retrieved; 98 contained valid sensor readings (10 heartbeat-only records excluded). Average temperature was 27.0°C (range: 22.5 – 30.0°C). Relative humidity averaged 69.7%. CO , averaged 415.6 ppm. Soil moisture averaged 30.6%. Water level averaged 6.7%.

### TEMPERATURE

27.0

!° 22.5 · avg · !° 30.0

### HUMIDITY

69.7

!° 56.9 · avg · !° 85.4

### CO ,

415.6

!° 300.0 · avg · !° 697.0

### SOIL MOISTURE

30.6

!° 0.0 · avg · !° 79.0

### WATER LEVEL

6.7

!° 0.0 · avg · !° 29.0

Records: 98

| Date Range: 3/28/2026 – 4/6/2026

| Params: Temperature · Humidity · CO , · Soil · Water

## 2. Detailed Sensor Data

| Timestamp              | Temp (°C) | Humidity (%) | CO , (ppm) | Soil (%) | Water (%) | Interval |
|------------------------|-----------|--------------|------------|----------|-----------|----------|
| 3/28/2026, 11:12:38 AM | 22.5°C    | 85.4%        | 547        | 69%      | 26%       | 180s     |
| 3/28/2026, 11:15:38 AM | 22.5°C    | 84.3%        | 568        | 74%      | 27%       | 180s     |
| 3/28/2026, 11:18:38 AM | 22.8°C    | 84.4%        | 585        | 0%       | 25%       | 180s     |
| 3/28/2026, 11:21:38 AM | 23.3°C    | 84.5%        | 612        | 65%      | 24%       | 180s     |
| 3/28/2026, 11:24:38 AM | 25.9°C    | 74.8%        | 626        | 67%      | 26%       | 180s     |
| 3/28/2026, 11:27:38 AM | 25.9°C    | 74.6%        | 637        | 66%      | 29%       | 180s     |
| 3/28/2026, 11:30:38 AM | 27.1°C    | 75.5%        | 652        | 65%      | 25%       | 180s     |
| 3/28/2026, 11:33:39 AM | 25.7°C    | 74.6%        | 673        | 67%      | 25%       | 180s     |
| 3/28/2026, 11:36:44 AM | 24.8°C    | 78.9%        | 697        | 70%      | 24%       | 180s     |
| 3/28/2026, 11:39:39 AM | 24.4°C    | 79.9%        | 642        | 70%      | 0%        | 180s     |
| 3/28/2026, 11:42:38 AM | 24.3°C    | 80.7%        | 593        | 71%      | 0%        | 180s     |
| 3/28/2026, 11:45:38 AM | 24.1°C    | 80.4%        | 593        | 70%      | 0%        | 180s     |
| 3/28/2026, 11:48:38 AM | 24.2°C    | 81.4%        | 596        | 71%      | 0%        | 180s     |
| 3/28/2026, 11:51:38 AM | 24.3°C    | 81.4%        | 592        | 72%      | 0%        | 180s     |
| 3/28/2026, 11:54:38 AM | 24.3°C    | 81.1%        | 602        | 72%      | 0%        | 180s     |
| 3/28/2026, 11:57:38 AM | 24.3°C    | 81.3%        | 616        | 73%      | 0%        | 180s     |

| Timestamp              | Temp (°C) | Humidity (%) | CO , (ppm) | Soil (%) | Water (%) | Interval |
|------------------------|-----------|--------------|------------|----------|-----------|----------|
| 3/28/2026, 12:00:38 PM | 24.3°C    | 80.8%        | 627        | 75%      | 0%        | 180s     |
| 3/28/2026, 12:03:38 PM | 24.4°C    | 80.3%        | 646        | 74%      | 0%        | 180s     |
| 3/28/2026, 12:06:38 PM | 24.4°C    | 81.8%        | 649        | 78%      | 29%       | 180s     |
| 3/28/2026, 12:09:38 PM | 24.4°C    | 82.0%        | 630        | 0%       | 24%       | 180s     |
| 3/28/2026, 12:12:38 PM | 24.7°C    | 80.7%        | 610        | 70%      | 22%       | 180s     |
| 3/28/2026, 12:15:38 PM | 24.7°C    | 80.7%        | 592        | 71%      | 23%       | 180s     |
| 3/28/2026, 12:18:38 PM | 24.8°C    | 79.7%        | 578        | 72%      | 19%       | 180s     |
| 3/28/2026, 12:21:38 PM | 24.9°C    | 79.2%        | 556        | 74%      | 19%       | 180s     |
| 3/28/2026, 12:24:38 PM | 25.1°C    | 79.7%        | 540        | 74%      | 19%       | 180s     |
| 3/28/2026, 12:27:38 PM | 25.2°C    | 78.5%        | 523        | 74%      | 20%       | 180s     |
| 3/28/2026, 12:30:39 PM | 25.4°C    | 79.2%        | 507        | 79%      | 22%       | 180s     |
| 3/28/2026, 12:33:38 PM | 25.5°C    | 78.6%        | 481        | 73%      | 22%       | 180s     |
| 3/28/2026, 12:36:38 PM | 25.5°C    | 78.4%        | 458        | 76%      | 22%       | 180s     |
| 3/28/2026, 12:39:38 PM | 25.6°C    | 78.2%        | 459        | 75%      | 22%       | 180s     |
| 3/28/2026, 12:42:38 PM | 25.7°C    | 76.5%        | 461        | 73%      | 22%       | 180s     |
| 3/28/2026, 12:45:38 PM | 25.7°C    | 75.6%        | 449        | 78%      | 22%       | 180s     |
| 3/28/2026, 12:48:38 PM | 25.9°C    | 75.4%        | 444        | 78%      | 22%       | 180s     |
| 3/28/2026, 12:51:38 PM | 26.0°C    | 76.3%        | 443        | 77%      | 22%       | 180s     |
| 3/28/2026, 12:54:38 PM | 25.7°C    | 76.4%        | 439        | 77%      | 23%       | 180s     |
| 3/28/2026, 12:57:38 PM | 25.6°C    | 76.9%        | 430        | 78%      | 23%       | 180s     |
| 3/28/2026, 1:00:38 PM  | 25.5°C    | 76.8%        | 423        | 78%      | 0%        | 180s     |
| 3/28/2026, 1:03:38 PM  | 25.4°C    | 77.8%        | 415        | 78%      | 5%        | 180s     |
| 3/28/2026, 1:06:38 PM  | 25.4°C    | 77.5%        | 415        | 78%      | 5%        | 180s     |
| 3/28/2026, 1:09:38 PM  | 25.4°C    | 77.6%        | 419        | 75%      | 5%        | 180s     |
| 3/28/2026, 1:12:38 PM  | 25.4°C    | 76.9%        | 423        | 75%      | 3%        | 180s     |
| 3/28/2026, 1:15:38 PM  | 25.4°C    | 76.6%        | 436        | 76%      | 6%        | 180s     |
| 3/28/2026, 1:18:38 PM  | 25.4°C    | 76.5%        | 443        | 75%      | 6%        | 180s     |
| 3/31/2026, 11:41:07 AM | 25.2°C    | 71.5%        | 390        | 0%       | 0%        | 180s     |
| 3/31/2026, 11:45:34 AM | 26.4°C    | 64.1%        | 370        | 0%       | 0%        | 220s     |
| 3/31/2026, 11:49:14 AM | 26.8°C    | 61.7%        | 360        | 0%       | 0%        | 220s     |
| 3/31/2026, 11:52:54 AM | 27.1°C    | 61.9%        | 350        | 0%       | 0%        | 220s     |
| 4/6/2026, 12:17:11 PM  | 27.3°C    | 69.4%        | 390        | 0%       | 0%        | 220s     |
| 4/6/2026, 12:20:51 PM  | 28.1°C    | 61.9%        | 380        | 0%       | 0%        | 220s     |
| 4/6/2026, 12:22:14 PM  | 28.5°C    | 60.8%        | 370        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:27:10 PM  | 28.7°C    | 60.9%        | 360        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:27:14 PM  | 28.7°C    | 60.9%        | 360        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:32:10 PM  | 29.1°C    | 59.8%        | 350        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:32:14 PM  | 29.1°C    | 59.8%        | 350        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:37:10 PM  | 29.3°C    | 57.8%        | 340        | 0%       | 0%        | 300s     |

| Timestamp             | Temp (°C) | Humidity (%) | CO , (ppm) | Soil (%) | Water (%) | Interval |
|-----------------------|-----------|--------------|------------|----------|-----------|----------|
| 4/6/2026, 12:37:14 PM | 29.3°C    | 57.8%        | 340        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:42:10 PM | 29.4°C    | 58.6%        | 330        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:42:14 PM | 29.4°C    | 58.6%        | 330        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:47:14 PM | 29.1°C    | 58.2%        | 320        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:52:14 PM | 28.7°C    | 58.4%        | 310        | 0%       | 0%        | 300s     |
| 4/6/2026, 12:57:14 PM | 28.9°C    | 58.7%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:02:14 PM  | 28.8°C    | 56.9%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:07:14 PM  | 28.4°C    | 58.5%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:12:14 PM  | 28.1°C    | 59.2%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:17:14 PM  | 28.4°C    | 60.5%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:22:14 PM  | 29.1°C    | 59.6%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:27:14 PM  | 29.7°C    | 59.3%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:32:14 PM  | 30.0°C    | 59.8%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:42:14 PM  | 29.5°C    | 62.0%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:47:14 PM  | 28.8°C    | 61.5%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:52:14 PM  | 28.4°C    | 62.8%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:57:14 PM  | 28.4°C    | 64.1%        | 300        | 0%       | 0%        | 300s     |
| 4/6/2026, 1:59:05 PM  | 28.6°C    | 64.3%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:00:05 PM  | 28.5°C    | 64.3%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:01:05 PM  | 28.6°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:02:05 PM  | 28.6°C    | 64.2%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:03:05 PM  | 28.6°C    | 64.3%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:04:05 PM  | 28.6°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:05:05 PM  | 28.6°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:06:05 PM  | 28.7°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:07:05 PM  | 28.7°C    | 64.5%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:08:05 PM  | 28.7°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:09:05 PM  | 28.7°C    | 64.3%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:10:05 PM  | 28.8°C    | 64.2%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:11:05 PM  | 28.9°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:12:05 PM  | 28.9°C    | 63.8%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:14:05 PM  | 28.9°C    | 63.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:15:05 PM  | 28.8°C    | 63.9%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:16:05 PM  | 28.8°C    | 64.2%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:17:05 PM  | 28.7°C    | 63.9%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:18:05 PM  | 28.7°C    | 63.8%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:19:05 PM  | 28.7°C    | 63.4%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:20:05 PM  | 28.7°C    | 63.7%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:21:05 PM  | 28.7°C    | 64.0%        | 300        | 0%       | 0%        | 60s      |

| Timestamp            | Temp (°C) | Humidity (%) | CO , (ppm) | Soil (%) | Water (%) | Interval |
|----------------------|-----------|--------------|------------|----------|-----------|----------|
| 4/6/2026, 2:22:05 PM | 28.7°C    | 64.1%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:23:05 PM | 28.6°C    | 64.5%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:24:05 PM | 28.6°C    | 64.7%        | 300        | 0%       | 0%        | 60s      |
| 4/6/2026, 2:25:05 PM | 28.6°C    | 64.4%        | 300        | 0%       | 0%        | 60s      |

### 3. Statistical Analysis

This section summarises statistical analysis across all monitored environmental parameters. Temperature ranged from 22.5 to 30.0 °C (avg 27.0 °C, <0 = 2.0). Humidity ranged from 56.9 to 85.4 % (avg 69.7 %, <0 = 8.7). CO , ranged from 300.0 to 697.0 ppm (avg 415.6 ppm, <0 = 127.8). Soil Moisture ranged from 0.0 to 79.0 % (avg 30.6 %, <2 Ò 3bã""à Water Level ranged from 0.0 to 29.0 % (avg 6.7 %, <2 Ò ãB' à

#### TEMPERATURE

| MIN     | MAX     | AVERAGE | STD DEV | SAMPLES |
|---------|---------|---------|---------|---------|
| 22.5 °C | 30.0 °C | 27.0 °C | 2.0 °C  | 98      |

#### HUMIDITY

| MIN    | MAX    | AVERAGE | STD DEV | SAMPLES |
|--------|--------|---------|---------|---------|
| 56.9 % | 85.4 % | 69.7 %  | 8.7 %   | 98      |

#### CO ,

| MIN       | MAX       | AVERAGE   | STD DEV   | SAMPLES |
|-----------|-----------|-----------|-----------|---------|
| 300.0 ppm | 697.0 ppm | 415.6 ppm | 127.8 ppm | 98      |

#### SOIL MOISTURE

| MIN   | MAX    | AVERAGE | STD DEV | SAMPLES |
|-------|--------|---------|---------|---------|
| 0.0 % | 79.0 % | 30.6 %  | 36.2 %  | 98      |

#### WATER LEVEL

| MIN   | MAX    | AVERAGE | STD DEV | SAMPLES |
|-------|--------|---------|---------|---------|
| 0.0 % | 29.0 % | 6.7 %   | 10.4 %  | 98      |

## 4. Data Visualizations

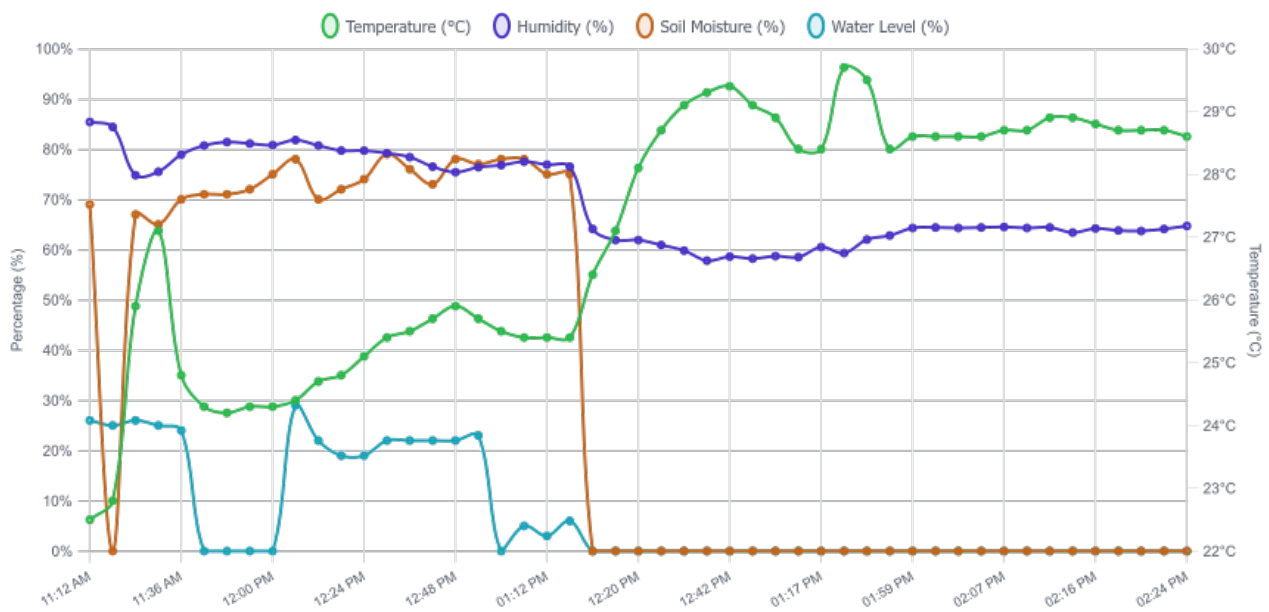


Figure 1: Environmental parameter trends over time (temperature on right axis, percentages on left axis)



Figure 2: CO<sub>2</sub> level distribution across all recorded readings

## 5. Recommendations

Based on collected data, the following actions are recommended:

- Temperature is within the optimal 18–28°C range. Continue regular monitoring.
- Humidity levels are within the recommended 30–75% range.
- CO<sub>2</sub> levels are within acceptable limits (< 800 ppm).
- Soil moisture levels are within the healthy 20–80% range.
- Water level is critically low (< 20%). Refill reservoir promptly.

**DATA QUALITY NOTE**

10 of 108 records were excluded (device heartbeats with no sensor readings).